YOLO Detection TENSORRT (TRT) Model Inference Report

Backend Engine: TensorRT (TRT)

Phase 1: Configuration & System Metadata

Model Name	yolo26n.engine
Input Dimensions	640x640
Source Images Directory	/workspace/Images
Processed Images Output	/workspace/ENGINE_Output
OS	Ubuntu 24.04.4 LTS
Python Version	3.12.3
TensorRT Version	10.16.1.11
Inference Device	NVIDIA RTX A4000
CUDA Version	13.2
Total Parameters	2,400,160
File Size	11.63 MB
Number of Classes	80
Precision Mode	FP32

Model Trained Class Names:

person, bicycle, car, motorcycle, airplane, bus, train, truck, boat, traffic light, fire hydrant, stop sign, parking meter, bench, bird, cat, dog, horse, sheep, cow, elephant, bear, zebra, giraffe, backpack, umbrella, handbag, tie, suitcase, frisbee, skis, snowboard, sports ball, kite, baseball bat, baseball glove, skateboard, surfboard, tennis racket, bottle, wine glass, cup, fork, knife, spoon, bowl, banana, apple, sandwich, orange, broccoli, carrot, hot dog, pizza, donut, cake, chair, couch, potted plant, bed, dining table, toilet, tv, laptop, mouse, remote, keyboard, cell phone, microwave, oven, toaster, sink, refrigerator, book, clock, vase, scissors, teddy bear, hair drier, toothbrush

Phase 2: Inference Performance Metrics

Total Images Processed	5
Preprocess Latency	68.61 ms per image
Postprocess Latency	115.88 ms per image
Peak Memory Usage	508.00 MB
Inference Latency (Without Warmup)	4.66 ms per image
Inference Latency (With Warmup)	1.68 ms per image
Throughput (Without Warmup)	5.29 Images/sec
Throughput (With Warmup)	5.37 Images/sec
Total Run Time (Without Warmup)	945.75 ms
Total Run Time (With Warmup)	930.85 ms

Phase 3: Detailed Predictions Summary

Image File	Detected Objects (Confidence)	Bounding Boxes (xyxy)
Cat_November_2010-1a.jpg.webp	cat (93.5%)	[208, 142, 892, 1219]
images.jpeg	dog (94.3%)	[87, 148, 422, 513]
images1.jpeg	zebra (92.6%)	[30, 173, 587, 416]
istockphoto-1189925691-612x612.jpg	bus (89.4%)	[69, 81, 583, 350]
tigermain.jpg	zebra (95.0%)	[668, 421, 1874, 1091]